VAYUSETU: AI Model Registry

Ensemble Weights, Hyperparameters, & Registered Checkpoints

1. Multi-Model AI Ensemble Strategy

VAYUSETU utilizes a multi-model ensemble blending spatial, temporal, decision tree, and physical models to produce highly reliable climate predictions. Rather than relying on a single algorithm, the ensemble fuses the outputs using a weighted combination script:

- **Convolutional LSTM (32%):** Captures spatial-temporal patterns, especially useful for formatting cloud movement and gridded precipitation forecasts.
- **Temporal Fusion Transformer (28%):** Captures long-term multi-horizon dependencies and temporal dynamics of temperature and humidity anomalies.
- **XGBoost Regressor (20%):** Models non-linear local features, extracting local geographic elevations and land surface anomalies.
- **Physics-Informed Neural Network (20%):** Constrains predictions to obey mass conservation, water budget equations, and thermodynamic thresholds (e.g. limiting evaporation relative to net solar radiation).

2. Registered Checkpoint Metadata

Model ID	Checkpoint Path	Input Tensor Shape	Parameters	Last Registered
convlstm_v1	models/convlstm_v1.pth	[Batch, 5, 1, 32, 32]	450k weights	2026-06-21 12:45
transformer_v1	models/transformer_v1.pth	[Batch, 5, 9]	820k weights	2026-06-21 12:46
xgboost_v1	models/ensemble_v1.pkl	[1, 14] vector	250 trees	2026-06-21 12:48
pinn_v1	models/pinn_v1.pth	[Batch, 4] physics parameters	120k weights	2026-06-21 12:50

3. Retraining and Model Drift Audits

A continuous drift auditor monitors prediction accuracy against ground observations using a Kolmogorov-Smirnov (KS) test. If the p-value falls below 0.05 (indicating significant statistical distribution shift), a 'Retrain Recommended' trigger is logged in the system registry, allowing operators to rebuild model layers with the latest assimilated twin states.